

1. Administrative & Study Identification

Official Study Title: An Open-Label Study to Investigate the Safety, Tolerability, and Pharmacokinetics/Pharmacodynamics of Risdiplam (R07034067) in Adult and Pediatric Patients With Spinal Muscular Atrophy **Short Title/Acronym:** JEWELFISH **Protocol Number:** BP39054 **NCT Number:** NCT03032172 **Sponsor Name:** Roche (Hoffmann-La Roche) **Investigational Product:** Risdiplam (R07034067 / EVRYSDI) **Phase of Development:** Phase 2 **Trial Status:** COMPLETED **Medical Monitor:** Hoffmann-La Roche Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To investigate the safety, tolerability, pharmacokinetics (PK), and PK/pharmacodynamic (PD) relationship of risdiplam in adult, children, and infant patients with Spinal Muscular Atrophy (SMA) previously treated with other SMA therapies. **Secondary Objectives:** To evaluate the Survival of Motor Neuron (SMN) protein levels in blood and conduct exploratory efficacy assessments of motor function.

2.2 Background & Rationale Spinal muscular atrophy (SMA) is a rare genetic disorder caused by mutations in the survival of motor neuron 1 (SMN1) gene, leading to low levels of functional SMN protein. Risdiplam is an orally administered, centrally and peripherally distributed SMN2 pre-mRNA splicing modifier. The JEWELFISH study was initiated to address the lack of data in previously treated patients, investigating the safety and PK/PD profile of risdiplam in individuals who had previously received treatments like nusinersen, olesoxime, AVXS-101, or the splicing modifier R06885247 (Moonfish study).

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm (Non-comparative) **Blinding:** Open-label **Randomization:** N/A (Exploratory)

3.2 Planned Enrollment & Duration Target \$N\$: 174 participants **Number of Sites:** Multicenter **Estimated Study Start:** 03-Mar-2017 **Estimated Primary Completion:** After 24 months of primary treatment phase

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Confirmed diagnosis of 5q-autosomal recessive SMA. 2. Previous enrollment in Study BP29420 (Moonfish) with the splicing modifier R06885247, OR previous treatment with one of the following: Nusinersen (≥ 4 doses, last dose ≥ 90 days prior to screening), Olesoxime (last dose ≤ 12 months and ≥ 90 days prior), or AVXS-101 (≥ 12 months prior). 3. Adequately recovered from any acute illness at screening and considered well enough to participate. 4. Women of childbearing potential: negative blood pregnancy test at screening, agreement to remain abstinent or use contraceptive measures, and agreement to refrain from donating eggs for at least 28 days after the final dose. 5. Men: agreement to remain abstinent or use contraceptive measures, and agreement to refrain from donating sperm. 6. Participants aged ≤ 2 years: Parent/caregiver willing to consider feeding tube placement (nasogastric, naso-jejunal, gastrostomy) to maintain hydration/nutrition/treatment delivery, and consider non-invasive ventilation if recommended.

4.2 Exclusion Criteria 1. Inability to meet study requirements or concomitant participation in any investigational drug/device study. 2. Previous participation in any investigational drug/device study within 90 days or 5 half-lives prior to screening (except olesoxime, AVXS-101, or nusinersen). 3. History of gene or cell therapy (with the exception of AVXS-101). 4. Unstable gastrointestinal, renal, hepatic, endocrine, or cardiovascular system diseases; or clinically significant ECG abnormalities/cardiovascular disease. 5. Inadequate venous or capillary blood access for study procedures. 6. Patients < 2 years: hospitalization for a pulmonary event within 2 months prior to screening and not fully recovered. 7. Lactating women. 8. Suspicion of regular consumption of drugs of abuse, or (for adults/adolescents) a positive urine test for drugs of abuse or alcohol at screening/Day -1. 9. History of malignancy if not considered cured. 10. Participants > 6 years: significant risk for suicidal behavior assessed by the Columbia-Suicide Severity Rating Scale (C-SSRS). 11. Any major illness within one month prior, or febrile illness within one week prior to screening up to first dose. 12. Recently initiated treatment (< 6 weeks prior) with oral salbutamol or another oral beta 2-adrenergic agonist. 13. Prior use of chloroquine, hydroxychloroquine, retigabin, vigabatrin, thioridazine, or inhibitors/inducers of FM01 or FM03 taken within 2 weeks (or 5 elimination half-lives) prior to dosing. 14. Ascertained or presumptive hypersensitivity to risdiplam or its constituents. 15. Recent history (< 1 year) of ophthalmological diseases.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Administered once daily. **Route of Administration:** Oral (liquid medicine). **Duration**

of Treatment: 24 months, followed by an optional open-label extension (OLE) phase for 3 years.

5.2 Concomitant & Prohibited Therapy Permitted: Standard of care for SMA, including routine physical rehabilitation and nutritional care. **Prohibited:** Concurrent use of other investigational SMA therapies, SMN2 splicing modifiers, or gene therapies during the risdiplam treatment period.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Genetic confirmation, Tanner Staging, Baseline Assessments
Baseline	Day 0	First Dose, PK/PD Labs, Motor Function Assessments
Interim	Months 6, 12, 18	Safety Labs, PK/PD tracking, Motor Function Assessments
End of Study	Month 24	Final Vital Signs, Efficacy Assessment, Option to enter Open-Label Extension (OLE)

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints

- 1. Number of Participants With Adverse Events (AEs):** Assessment of any untoward medical occurrences regardless of causal attribution.
- 2. Number of Participants Who Discontinued Treatment Due to AEs:** Tracking of participants who withdrew from risdiplam treatment as a result of an adverse event.
- 3. Number of Participants With Shift in Puberty Status From Baseline:** Measured via Tanner staging of sexual development (Stages I to V) across expected developmental timelines for participants.

7.2 Secondary & Exploratory Endpoints

Secondary Outcome: Survival of Motor Neuron (SMN) Protein Levels in Blood.

Exploratory Efficacy: Stabilization or changes in total scores assessed by the 32-item Motor Function Measure (MFM32), Hammersmith Functional Motor Scale-Expanded (HFMSE), and Revised Upper Limb Module (RULM).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring via patient/caregiver reporting and clinical evaluation at onsite visits.

Serious Adverse Events (SAEs): Standard reporting timeline (typically 24 hours). The most common SAE noted in the study was pneumonia.

Data Monitoring Committee (DMC): An independent Data Monitoring Committee monitored the risdiplam studies on an ongoing basis.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set (all participants who received at least one dose of risdiplam) and PK/PD analysis sets. PK data was analyzed in a single arm group irrespective of prior therapies. **Sample Size Justification:** N=174. The study was primarily designed to assess safety, PK, and PD; it was not powered for formal efficacy analysis. **Handling of Missing Data:** Descriptive statistics applied without formal imputation for exploratory efficacy endpoints. Pubertal development data analyzed across Normal, Delayed, and Missing categories.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study was conducted in accordance with Good Clinical Practice (GCP) guidelines and the principles of the Declaration of Helsinki. **Monitoring Plan:** Regular onsite monitoring visits supplemented by remote safety check-ins. **Audit Trail:** Standard institutional review board approvals and eCRF data clarification processes.

11. Study Identifiers & Governance

Primary ID: BP39054 **Registry Number:** NCT03032172 **Sponsor/Lead Organization:** Roche (Hoffmann-La Roche) **Study Title:** JEWELFISH **Official Regulatory Title:** An Open-Label Study to Investigate the Safety, Tolerability, and Pharmacokinetics/Pharmacodynamics of Risdiplam (R07034067) in Adult and Pediatric Patients With Spinal Muscular Atrophy

12. Clinical Framework (SMA Specific)

Primary Condition: Spinal Muscular Atrophy (SMA) **Diagnosis Threshold:** Confirmed 5q-autosomal recessive SMA diagnosis **Medical Methodology:** Survival of motor neuron 2 (SMN2) pre-mRNA splicing modifier therapy **Optimization Target:** Increased production of functional SMN protein

13. Study Procedures & Methodology

Management Pathway: Standard clinical pathway for SMA management **Education Protocol (HCP):** Training on risdiplam dosing calculations and safety monitoring **Education Protocol (Patient):** Onsite training/self-learning regarding daily oral liquid administration at home **Quality Audit Mechanism:** Adherence monitoring, continuous AE tracking, and routine motor function assessment evaluations

14. Observation & Follow-Up Schedule

Total Duration: 24 months of primary treatment + 3 years Open-Label Extension **Onsite Visit Cadence:** Baseline, Months 6, 12, 18, and 24 **Remote Monitoring Frequency:** Routine safety follow-ups as defined by standard clinical practice **Checklist Review Interval:** Every 6 months alongside major clinical assessments

15. Sign-off & Approval

Role	Name	Signature	Date		----	----	----	----
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]					Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]					